

Problem 8.13. Define $A_{\text{LBA}} = \{\langle M, w \rangle \mid M \text{ is an LBA that accepts input } w\}$. Show that A_{LBA} is PSPACE-complete.

Proof. To show that A_{LBA} is PSPACE-complete, we must show that it is in PSPACE and that all PSPACE problems are polynomial time reducible to it. The algorithm given in Theorem 5 decides A_{LBA} ¹. This algorithm runs in polynomial space and so $A_{\text{LBA}} \in \text{PSPACE}$. To prove that A_{LBA} is PSPACE-hard, we give a polynomial time reduction from GG to A_{LBA} . This reduction converts a generalized geography graph G to an LBA L and a string w so that Player I has a winning strategy for the game played on G , iff L accepts w .

Part (b) Notes:

Show GG is polynomial time reducible to A_{LBA} . Rewrite PSPACE decider for GG to use a stack. The mapping f takes a directed graph and outputs an LBA, graph and enough blank space for stack. Size of stack is same as number of nodes in graph.

□

¹An LBA operates in exactly n space. This solution applies to both DLBA and NLBA due to Savitch's Theorem.